Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

WAVE Thematic Assembly Centre:

WAVE_GLO_PHY_SPC-FWK_L3_NRT_014_002

cmems-obs-wave_glo_phy_spc-fwk_nrt_s1a_l3_PT15S-i

cmems-obs-wave_glo_phy_spc-fwk_nrt_s1b_l3_PT15S-i

cmems-obs-wave_glo_phy_spc-fwk_nrt_cfo_l3_PT15S-i

Issue: 1.5

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Approval date by the CMEMS product quality coordination team: 17/11/2025

CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table. The third column specifies which Sections or Sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	June2018	All	First version of document at CMEMS V4, done from previous Wave-TAC QUID for the SAR wave component	R Husson, G. Goimard	Mercator Océan
1.1	26.04. 2019		Feedbacks from Mercator on Quid + New processing to improve long swell description	R Husson	
1.1	09.07. 2019		Feedbacks from Mercator on Quid	R Husson	Mercator Ocean
1.2	03.04. 2020		Addition of Quality Flag from L2 swell partitions	R Husson	
1.3	10.09. 2021		EIS of the CFOSAT/SWIM L3 SPC products	R Husson	Mercator Ocean
1.4	10.10. 2024		Indicating the end of mission for S1B Rename of product and datasets	P. Zunino	P. Zunino
1.5	21.05.2025		Enhancement of filtering of CFOSAT input data. Renaming of dataset	M. De Carlo	P. Zunino

ACRONYM TABLE

Acronym	Description
CFOSAT	China-France Oceanography SATetille
CLS	Collecte Localisation Satellite
CNES	Centre National d'Etudes Spatiales
ECMWF	European Center for Medium-Range Weather Forecasts
EIS	Entry Into Service
ESA	European Space Agency
FWK	Fireworks
Hs	Significant wave height (referring to the wave parameter)
L1	Level-1
L2	Level-2
L2 OCN	Level-2 OCeaN
L2 OSW	Level-2 Ocean SWell
L2P	Level-2 Plus
L3	Level-3
L4	Level-4
MY	Multi-year
NDBC	National Data Buoy Center
NetCDF	network Common Data Form
NOAA	National Oceanic and Atmospheric Administration
NRT	Near Real Time
OSISAF	Ocean and Sea Ice Satellite Application Facility
PCC	Pearson Correlation Coefficient
PUM	Product User Manual
QUID	Quality information Document
REP	Reprocessing
RMSD	Root-mean Square Deviation
S1	Sentinel-1 (ESA's serie of SAR satellite)

S1A	Sentinel-1 A
S1B	Sentinel-1 B
SAR	Synthetic Aperture Radar
SI	Scatter Index
SPC	Spectral
SWH	Significant Wave Height (referring to products that only contain the Hs parameter)
SWIM	Surface Wave Investigation and Monitoring
TAC	Copernicus Marine Service Thematic Assembly Centre
USA	United States of America
VAVH	Visual Average Height (variable name associated to Significant Wave Height)
VPED	Peak direction of wave system (variable name in file)
VTPK	Peak period of wave system (variable name in file)

KEY DEFINITIONS

	Description
Date format	Throughout this document, the format for dates is DD.MM.YYYY
Fireworks	name of the propagation and storm refocusing algorithm described in Husson, 2012
Level-2 Plus	for CFOSAT data: similar to a L2 with some additional information (such as flags)
Level-2 OCeaN	for Sentinel 1 data: L2 with the particular Ocean process applied, includes information with the OSW process
Level-2 Ocean SWell	for Sentinel 1 data: part of the L2 OCN product with only the information pertaining to swell
Pearson correlation coefficient	In statistics, the Pearson correlation coefficient (PCC)[a] is a correlation coefficient that measures linear correlation between two sets of data. It is the ratio between the covariance of two variables and the product of their standard deviations. [from Wikipedia Pearson correlation coefficient]
Q-Q plot	Plot to compare 2 probability distributions by plotting their quantiles against each other. A point (x, y) on the plot corresponds to one of the quantiles of the second distribution (y-coordinate) plotted against the same quantile of the first distribution (x-coordinate). [from Wikipedia Q-Q plot]
SWIM	Payload onboard CFOSAT (the satellite), composed by a rotating antenna which allow to obtain wave spectra. For more information, see https://www.aviso.altimetry.fr/en/missions/current-missions/cfosat/instruments/swim.html
TREX	Name of CNES's computing facility (not an Acronym)

RELEVANT DOCUMENT LIST

Reference	Document
[PUM L3 SPC-FWK]	Product User Manual for WAVE TAC product WAVE_GLO_PHY_SPC-FWK_L3_NRT_014_002 https://documentation.marine.copernicus.eu/PUM/CMEMS-WAV-PUM-014_002.pdf
[CLSMonitor]	CLS Monitoring website http://satwave-report.cls.fr/
[S1-ATBD]	Sentinel-1 Ocean Swell Wave Spectra (OSW) ATBD, v1.9, 2025, September 8th - Sentinel Online. Available on https://sentiwiki.copernicus.eu/web/document-library#DocumentLibrary-SENTINEL-1DocumentsLibrary-S1-Documents

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the estimation process and the quality of the operational Near Real Time (NRT) WAVE_GLO_PHY_SPC-FWK_L3_NRT_014_002 (see Table 1 for main characteristics). This product is a Level-3 (L3) product containing waves spectral information over the global ocean. The waves spectral information helps characterize the "sea states" by providing the energy, peak direction and peak wavelength of the different wave systems present at a given location[1]. This product is a SPeCtral (SPC) product derived from an algorithm known as FireWorKs (FWK) [2] which propagates waves spectral information from satellite observations over the whole world using a linear propagation model. SPC products are differentiated from Significant Wave Height (SWH) products typically measured by altimeters as they also provide frequency and directional information.

This product is associated to three datasets, each associated with a satellite mission:

- "cmems-obs-wave_glo_phy_spc-fwk_nrt_s1a_l3_PT15S-i" for Sentinel-1A (S1A)
- "cmems-obs-wave_glo_phy_spc-fwk_nrt_s1b_l3_PT15S-i" for Sentinel-1B (S1B)
- "cmems-obs-wave_glo_phy_spc-fwk_nrt_cfo_l3_PT15S-i" for the China-France Oceanography SATetille (CFOSAT)

The Sentinel-1 satellites use C-band Synthetic Aperture Radar (SAR), while the CFOSAT satellite uses a Rotating Radar known as Surface Wave Investigation and Monitoring (SWIM).

This product is referred to as L3 SPC-FWK and is distributed in the form of NetCDF (network Common Data Form) files containing the satellite observations along with the 'pseudo-observations' which corresponds to the propagated observations. For each observation or 'pseudo-observation', it contains information on the three main integral parameters of the most energetic swell fields:

- significant wave height (Hs)
- peak period (Tp) or peak wavelength (Wl),
- peak direction (Dp).

[1] For more information about how to describe a sea state, see https://help.marine.copernicus.eu/en/articles/6175153-how-to-describe-the-wave-height-and-wave-period-parameters

[2] The name of the algorithm is due to the representation of the swell, propagating away from their generation area across an entire ocean basin (e.g. on , chapter IV 2.2.2).

Table 1 - Main characteristics of the L3 SPC-FWK NRT product

Product	WAVE_GLO_PHY_SPC-FWK_L3_NRT_014_002
Area	Global ocean
Missions	S1A; S1B; CFOSAT/SWIM
Main information	Swell partition integral parameters (significant wave height, peak period and peak direction) delivered along swell propagation track, from storm source until reaching land. Swell propagated observations related to the same swell field are stored in the same file.
Spatial resolution	Spatial resolution not applicable (as delivered along swell track and varying from one swell partition to another)
Temporal resolution	3-hour

The number of data processed by the system varies with time, according to satellites availability(see Table 2). It equals approximately 60,000 wave spectra by satellite. Table 2 summarizes the time periods during which the datasets for each mission are available.

Table 2: Temporal period processed by the L3 SPC wave chain for the different datasets. Those periods are necessarily shorter than L2 availability presented in Table 4.

satellite	Temporal availability	
	Begin date	End date
S1A	04.01.2020	Present
S1B	04.01.2020	23.12.2021
CFOSAT	01.12.2021-	Present

I.2 Summary of the results

The quality of the SPC FWK product is estimated using dynamical co-located crossovers [Wang et al. 2020, Wang et al. 2022]. In practice, propagated observations are co-located in space and time to compare the performances for different geometries (ascending/descending orbits), different incidence angles (WaVemode1 (WV1) vs WaVemode2 (WV2), corresponding to observations at 23° and 36° of incidence angle for Sentinel-1 and Left vs. Right for CFOSAT) and different satellites (S1A vs. S1B, S1A vs CFOSAT, S1B vs. CFOSAT).

The propagated observations are also co-located and compared to National Data Buoy Center (NDBC) buoys. The statistical analysis of the peak wavelength and peak direction is estimated at buoy location from the L3 Copernicus Marine Environment Monitoring Service (CMEMS) spectral products using a virtual observer and buoy measurements from the National Data Buoy Center (NDBC[3]) buoy network. This analysis assesses the accuracy of the measurements selected in the Level-3 product with respect to

[3] https://www.ndbc.noaa.gov/

the overall measurements, without any quality control. It is conducted by comparing propagated swell observations that are refocused (as detailed in section II.2.3.6) with respect to the unfocused propagated swell observations, referred to as left-overs.

The following paragraphs present the main results obtained.

Impact on selecting refocused data (comparisons between Sentinel-1 configurations):

With respect to the overall set of Level-2 (L2) swell observations used in the L3 analysis (i.e. which have passed the selection criteria detailed in Table 7), 14% of the L2 input swell observations are kept in the L3 CMEMS spectral products for internal comparison.

The biases remain approximately the same for the peak direction and significant wave height when comparing the inputs and the refocused propagated observations. The peak wavelength, however, shows a bias improvement from -14m to -1.3m with the refocused data.

The RMSD (Root Mean Square Difference) is greatly improved for both:

- From 98 m to 26 m for the peak wavelength,
- From 83° to 8° for the peak direction,

Comparison with in-situ data:

Overall, comparisons between SWIM L3 product and NDBC measurement and S1 L3 products and NDBC measurements, indicate:

- There are about 7 times more SWIM products co-located with NDBC measurements than S1A. This is partly due to the geographical coverage of the satellites and to the new data filtering applied to select only the most relevant observations.
- The scatter index for SWIM products is of the same order of magnitude as the one for S1A WV1 and S1A WV2, for all partition integral parameters.
- The significant bias for SWIM partition peak wavelength has been drastically reduced compared to previous version of the QUID. With a SWIM partition peak wavelength estimated from a slope spectrum there was an underestimation of 19 m compared with buoys and an underestimation of 30m compared with S1A. Now, using a peak wavelength estimated from a height spectrum the bias is of 2m for the buoy comparison and 16.8 m for the comparison with S1A.

I.3 Estimated Accuracy Numbers

The bias, Root-Mean Square Difference (RMSD) and Scatter Index (SI) for pairwise comparisons between S1A, CFOSAT and NDBC is shown in Table 3 for the three parameters of interest: the significant wave height, the peak wavelength and the peak direction. These numbers represent the global validation results.

Table 3: Estimated Accuracy Numbers (EANs) for Significant wave height (Hs), peak wavelength (Wl) and peak direction (Dp), computed from 01.01.2022 to 31.01.2022

	Metric names	Comparison between		Bias	RMSD	SI
		x-axis	y-axis	(m)	(m)	(%)
Hs	Hs_L3-NC-S1A-SWI-<metric>_T-XOVER	L3 from S1	L3 from CFOSAT	-0.02	0.59	45.22
	Hs_L3-CLASS2-MOOR-<metric>_S1A-WV2	NDBC buoys	L3 from S1A - WV2	0.16	0.71	43.54
	Hs_L3-CLASS2-MOOR-<metric>_SWI	NDBC buoys	L3 from CFOSAT	0.123	0.72	51.17
	Metric names	**Comparison between**		**Bias**	**RMSD**	**SI**
		x-axis	**y-axis**	**(m)**	**(m)**	**(%)**
Wl	Wl_L3-NC-S1A-SWI-<metric>_T-XOVER	L3 from S1A - WV2	L3 from CFOSAT	-16.8	38.02	11.26
	Wl_L3-CLASS2-MOOR-<metric>_S1A-WV2	NDBC buoys	L3 from S1A - WV2	14.98	50.50	15.95
	Wl_L3-CLASS2-MOOR-<metric>_SWI	NDBC buoys	L3 from CFOSAT	-4.39	60.28	19.62
	Metric name	**Comparison between**		**Bias**	**RMSD**	**SI**
		x-axis	**y-axis**	**(°)**	**(°)**	**(%)**
Dp	Dp_L3-NC-S1A-SWI-<metric>_T-XOVER	L3 from S1A - WV2	L3 from CFOSAT	-0.97	9.24	-13.21
	Dp_L3-CLASS2-MOOR-<metric>_S1A-WV2	NDBC buoys	L3 from S1A - WV2	-0.74	37.73	14.38
	Dp_L3-CLASS2-MOOR-<metric>_SWI	NDBC buoys	L3 from CFOSAT	-3.09	58.82	22.61

II PRODUCTION SYSTEM DESCRIPTION

The production centre name is WAVE-CLS-BREST-FR and the production system is referred to as the L3 SPeCtral-FireWorKs (SPC-FWK) wave chain.

II.1 Overview of the SAR-derived spectral measurements

The L3 spectral products are based on:

- L2 Wave Mode (WM) products from Sentinel-1A and Sentinel-1B and
- Level-2 Plus (L2P) products from CFOSAT/SWIM instrument.

These L2 and L2P products are measured directional ocean wave spectra.
For Sentinel-1, the ocean wave L2 products are also referred to as Ocean SWell (OSW). The OSW component provides continuous measurements of SAR swell spectra at C-band and are produced by the European Space Agency (ESA) ground segment. For SWIM, the L2P Products are produced by CLS on the Centre National d'Etudes Spatiales (CNES, i.e. French Spatial Agency) computing facility named "TREX".

Input wave spectra are illustrated in Figure 1and Figure 2Figure 2, with spectra taken in the Northern Pacific Ocean (ca 2,000 km north-west of Hawaii). They illustrate the wave energy distributed on a polar grid (k, phi), where k is the wavenumber and phi the angle relative to the North, and exhibit the respective limitations of the spectra:

- Long swell imaged by Sentinel-1 with no short wave seen in azimuth direction due to the azimuth cut-off, delimited by magenta dashed lines (see Figure 1),
- Waves seen in all directions by SWIM, but with limited capability to image long swell (>500m) and inability to remove the 180° propagation ambiguity (see Figure 2).

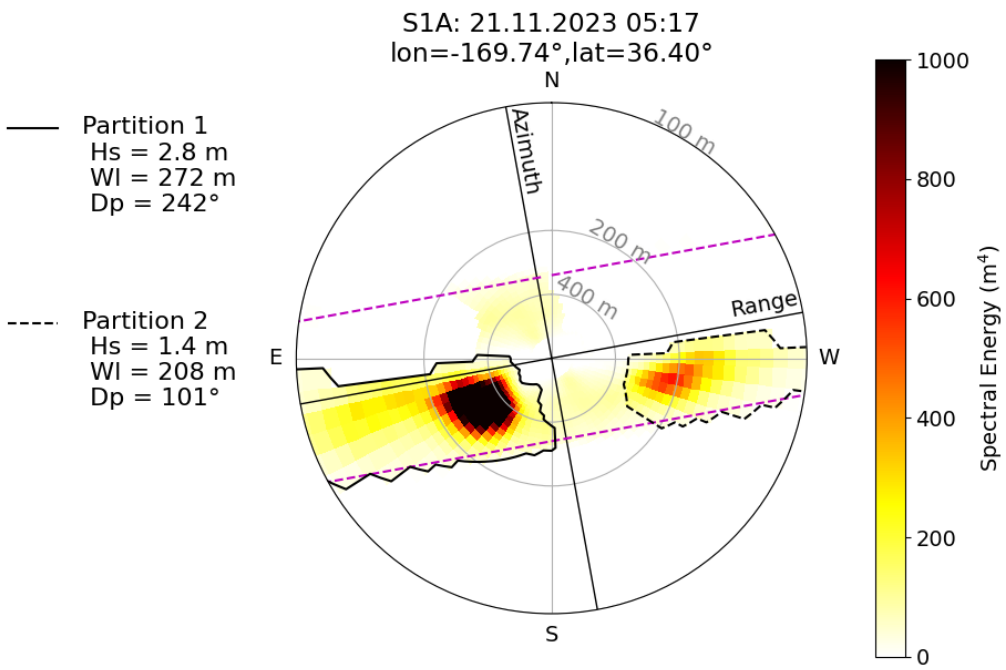

Figure 1: Example of a Level-2 (L2) Ocean SWell (OSW) product acquired by the Sentinel-1A (S1A) satellite, on the Northern Pacific Ocean. It depicts a directional wave spectrum containing two wave partitions, which are delineated in solid and dashed black lines and whose characteristics (peak wavelength (Wl), peak direction (Dp) and significant wave height (Hs)) are given beside the spectrum. The magenta dashed line corresponds to the azimuth cut-off of S1A

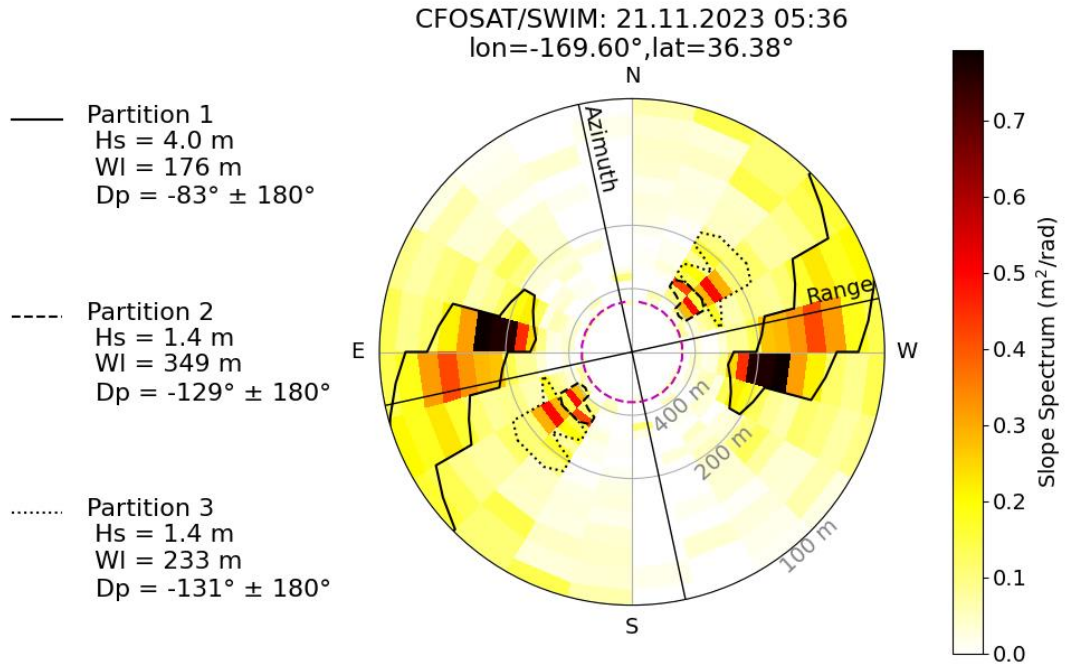

Figure 2: Example of a Level-2 Plus (L2P) product acquired by the CFOSAT satellite, on the Northern Pacific Ocean. It depicts a directional wave slope spectrum containing 3 wave partitions, which are delineated in solid, dashed and dotted black lines and whose characteristics (peak wavelength (Wl), peak direction (Dp) and significant wave height (Hs)) are given beside the spectrum. The magenta dashed line represents the 500-m wavelength cut-off of CFOSAT/SWIM

II.2 Production centre description for the version covered by this document

II.2.1 Spectral Fireworks (SPC-FWK) Wave chain overview

The system's primary objective is to provide operational products of quality flagged SPC data for Sentinel-1A, and CFOSAT missions, and historically Sentinel-1B.

Figure 3 illustrates the architecture in place for Near Real-Time (NRT) production. The details of the numbered elements are described below.

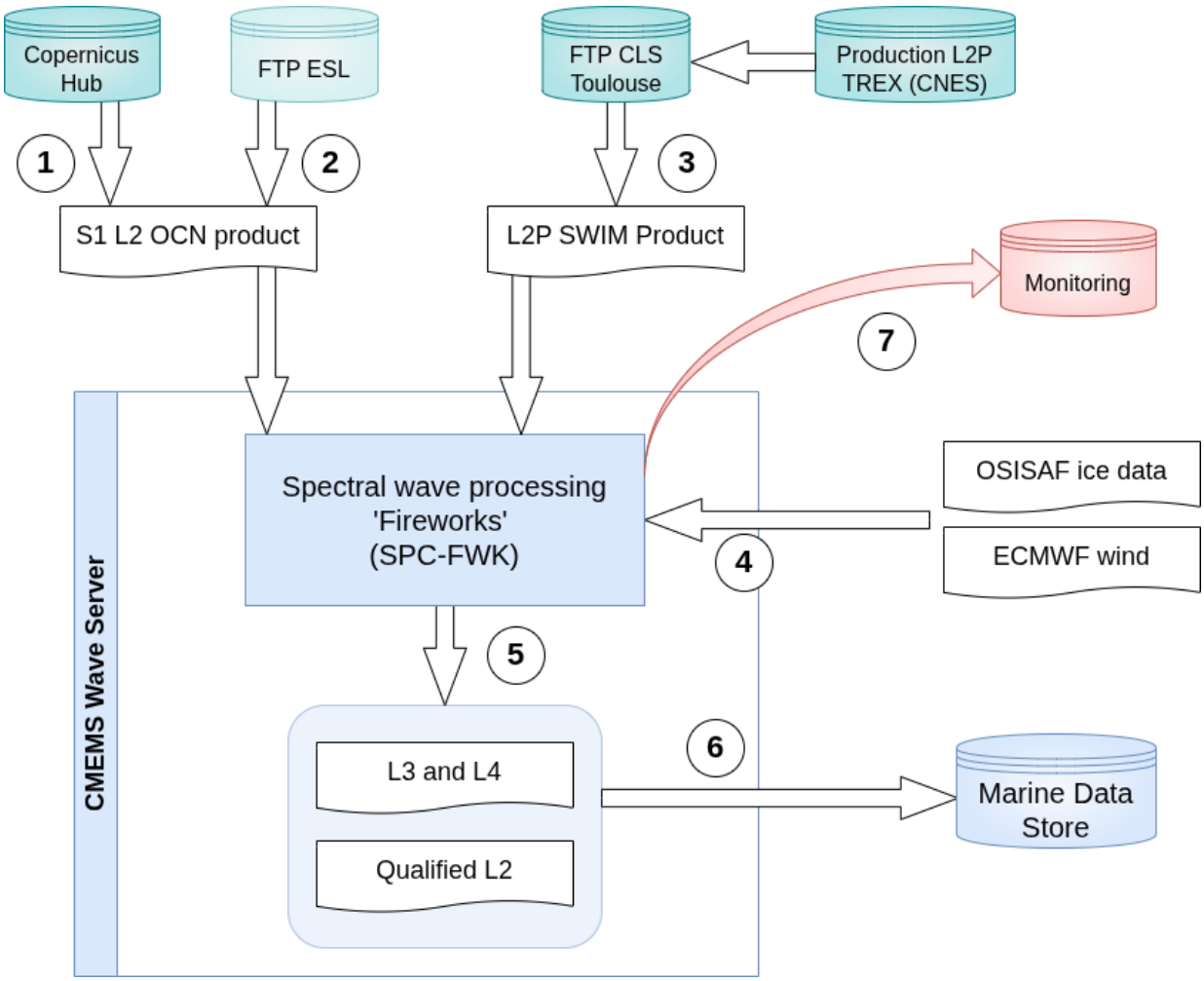

Figure 3: Architecture of the SPC-FWK chain in place for Near Real-Time (NRT). Further details on the specific steps (see numbers) can be found in the main text below. The data providers are the Copernicus service and the Expert Support Laboratory (ESL) for the European Spatial Agency (ESA) Sentinel-1 production of Level-2 OCeaN (L2 OCN) products; the French Centre National d'Etudes Spatiales (CNES) which computing facility 'TREX' is synchronised via FTP to CLS (Collecte Localisation Satellites, Copernicus Marine Service Wave operator) server for the CFOSAT/Swim production. Ancillary data are provided by the European Center for Medium range Weather Forecast (ECMWF) for the wind, and the Ocean and Sea Ice Satellite Application Facility (OSISAF) for the.

Figure 3 illustrates the architecture in place for Near Real-Time (NRT) production, while the steps followed by the SPC-FWK are detailed below:

Input acquisition:

1. The L2 OCN products , used as inputs, are gathered from the Copernicus Data Space Hub,
2. As a back-up, input data can be gathered from the MPC-S1 (Mission Performance Center for Sentinel-1) FTP (File Transfer Protocol) for Sentinel-1 products,
3. L2P SWIM data are downloaded from the CLS FTP synchronized with the Centre national d'études spatiales (CNES) computing facility called "TREX" which is the L2P production server.
4. OSISAF (Ocean and Sea Ice Satellite Application Facility) ice data and ECMWF (European Centre for Medium-Range Weather Forecasts) wind data are used in the calculation process to constrain the wave propagation and better define the swell generation regions,

Product generation:

5. L3 products are generated and stored,

Product dissemination:

6. After the calculation process, the results are sent through an ftp link on the Copernicus Marine Service Dissemination Unit Delivery interfaces.

Monitoring:

7. To ensure a good follow-up of the process, the server and the process are monitored with an event monitoring system such as Nagios[5].

II.2.2 Acquisition

The spectral measurements used in the system are the swell integral spectral parameters given in the L2 products from Sentinel-1A and Sentinel-1B missions and L2P products from the CFOSAT/SWIM instrument. Their source, delay and period of availability are summarised in Table 4. Mission characteristics are presented in Table 5. L2 and L2P availability periods are longer than L3-wave products introduced in the L3 WAVE-TAC system (see Table 2). Since their production started, L2 have been generated using several successive versions of the ESA IPF (Instrument Processor Facility). Similarly, L2P have been generated using several successive versions of the CNES CFOSAT/SWIM processor. Generating a homogeneous L3 dataset since the beginning of their availability period is thus not relevant before any input reprocessing is performed.

Table 4: Source, delay and period of availability of the different SAR data.

Mission	Type of product	Source	Availability delay	Period of availability
Sentinel-1A	L2 WV	ESA/CopHub	50% data in ~4h30	04.2014– present
Sentinel-1B	L2 WV	ESA/CopHub	50% data in ~4h30	04.2016 – 23.12.2021
CFOSAT	L2P	CLS/CNES HAL	Not evaluated yet	10.2018 - present

[5] see https://www.nagios.org/ or https://en.wikipedia.org/wiki/Nagios for more information

Table 5: SPC mission characteristics.

Mission	Cycle duration (days)	Number of orbits per cycle	Sun-synchronous	Acquisition mode
Sentinel-1A	12 (6 with both S1)	175	Yes	WV
Sentinel-1B	12 (6 with both S1)	175	Yes	WV
CFOSAT	13	197	Yes	10° incidence angle

The purpose of the acquisition processing is to acquire all the files necessary to compute the L3 products for each data source. This includes:

- The new L2 files,
- The contextual information used during the computation: Ice concentration and surface wind which come from OSISAF analysis and the ECMWF 0.125° model respectively.

II.2.3 L3 generation

II.2.3.1 Objective

This section presents the methodology through which isolated input spectral wave measurements are analyzed to produce the L3 CMEMS wave spectral products. The main motivation for proposing these files is twofold:

- Propose a quality flag for input products: This flagging is provided based on the overall consistency between a single spectral wave measurement and its neighbors in space and time. To be relevant, this comparison must be performed among the wave measurements related to the same swell field: it corresponds to the group of waves that have been generating by the same storm event. They can propagate for thousands of kilometers and show predictable variation in space and time [Munk et al. 1963].
- Predict the evolution in space and time of the swell observations: Based on a linear propagating wave theory, the swell field analysis performed on the input products enables estimation of the evolution before and after observation time of the wave integral parameters, derived from each wave partition extracted from the input products: significant wave height, peak period (or peak wavelength using the wave dispersion relation) and peak direction.

II.2.3.2 Overall methodology

A mandatory step to be able to perform these two tasks is to group the input wave measurements in swell fields, even though they are observed at various times and locations. Such observation of the same swell field at a global scale using satellite spectral wave measurements have been repeatedly reported in the past using several SAR imaging instruments operating in a specific wave mode, from ERS-1 to ENVISAT/ASAR (Holt et al., 1998; Heimbach and Hasselmann, 2000; Collard et al., 2009; Husson, 2012).

This association in swell fields assumes observations originate from the same source region. Inversely, storm regions are defined in our approach by the regions where a significant amount of back-traced swell observations converge.

The swell field reconstruction principle is rather simple: swell observations' trajectories are traced back in the past, and they converge to their storm source region. Then, these converging retro-propagated observations are associated. They define a new swell field. Its spatio-temporal properties are described by these observations, propagating from the storm source region until shallow water is encountered. The overall methodology can be split into the following steps:

1. **Data selection:** from all the input wave products, only a subset is evaluated. This data selection is further detailed in section II.2.3.4;

2. **Retro-propagation:** tracing the swell observations trajectories back in the past; by propagating the observations using a linear propagation model.

3. **Refocusing:** swell sources are defined by the regions where retro-propagated swell observations converge;

4. **Swell observation association:** all the retro-propagated observations within the previously detected refocusing region are gathered and define the new swell field. The generation time and region are the same for all observations;

5. **Swell field refined analysis:** generation time and region are re-estimated for each observation belonging to the new swell field. Swell observations can show different generation times and locations due to long-lasting or fast-moving storm events. The evolution of the significant wave height, which depends on the generation time, is defined from the storm source to land.

6. **Swell observations' filter:** the spatio-temporal consistency of the swell field is used to reject swell partitions that are judged inconsistent with the overall swell field based on the space/time structure of the integral parameter distribution.

7. **Swell field association:** in the end, previously detected swell fields that were split into separate refocusing events but describe the same event is gathered. It is based on a spectral distance criterion between two swell fields.

8. **Generation of the L3 products:** one NetCDF file is generated for each swell field, gathering spectral information on the propagated observations as well as on the newly qualified observations.

II.2.3.3 Background hypothesis

The main hypothesis on which the present method is based is the linear wave propagation theory. It is only applied in the deep ocean and assumes the absence of currents.

Only wave observations that can be back-traced following great circles (geodesics) to a common region are selected in the Level-3 products. This method enables the rejection of wave observations showing inconsistent peak direction and period that can be due to the degraded quality of input products. One side effect is also to reject wave observations that have been significantly affected by non-linear effects

such as diffraction or refraction effects due to bathymetry or currents which are not taken into account in our methodology.

II.2.3.4 Data selection

The data selection aims at rejecting wave observations which are not suitable for the Level-3 generation. The conditions and impact on selected dataset are presented in Table 6 and Table 7.

Table 6: Flags and thresholds criteria for data selection for the different missions.

Parameter	Sentinel-1A	Sentinel-1B	CFOSAT
Partition peak wavelength	Min: 200 m	Min: 200 m	Min: 200 m
Partition significant wave height	Min Hs: 0.30 m	Min Hs: 0.30 m	Min Hs: 0.3 m + continuity criterion between 0.3 and 1m
Observed latitude	-75 <Lat< 75	-75 <Lat< 75	-75 <Lat< 75
Land Flag	oswLandFlag = 0	oswLandFlag = 0	Already filtered in the L2P

Table 7: Percentage of rejected swell partitions estimated by S1-A and S1-B from WV1 and WV2 acquisition mode over week 23 – 2018 and by CFOSAT over week 2 - 2022. These weeks are chosen as representative examples.

Parameter	Sentinel-1A		Sentinel-1B		CFOSAT (v<1.5)	CFOSAT (v≥1.5)
Incidence Angle	WV1 (23°)	WV2(36°)	WV1 (23°)	WV2(36°)	10°	10°
Total	35651	42971	27064	34804	130269	130269
Partition peak wavelength	5625 15.8%	7600 17%	6801 25.1%	8872 25.5%	80288 61.6 %	80288 61.6%
Partition significant wave height	3728 10.4%	5794 13.5%	3250 12.0%	7056 20.3%	31504 24.2%	15720 12.1%
Observed Latitude	0 0%	0 0%	1 0%	3 0%	0 0.0%	0 0.0%
Wind speed	0 0%	0 0%	0 0%	0 0%	302 0.23%	302 0.23%
All previous criteria	9463 **26.5%**	15026 **35.0%**	8842 **32.7%**	12632 **36.3%**	**86.0%**	96310 73.9 %
oswLandFlag	>32.3%	>26.5%	>4.8%	>9.1%	0%	0%
All previous criteria	20972 **58.8%**	26419 **61.5%**	10156 **37.5%**	15796 **45.4%**	112094 **86.0%**	96310 73.9%

The condition on the minimum partition significant wavelength aims at selecting the longest waves, which better propagates over very large distances and better follow the hypothesis of linear wave

propagation. These long waves that can propagate for thousands of kilometres and are independent of the wind conditions are referred to as *swell*.

The condition on the minimum significant wave height (H_s) aims at rejecting swell observations within the estimation error of the partition (H_s).

Before the Nov 2025 Entry Into Service (EIS), the minimum H_s for CFOSAT observations was 1 m. Afterwards, the filtering of CFOSAT observations is modified to allow the selection of some observations with H_s between 0.3 m and 1 m. Due to parasitic peaks, not all CFOSAT observations are relevant for H_s below 1m, therefore a more complex selection is needed. An analysis is then performed over each satellite half-orbit pass: the along-track continuity of the partitions integral parameters is investigated to form clusters of consistent partitions (likely related to the same event). Partitions with H_s above 0.3 m and pertaining to such a cluster of at least five partitions are considered relevant and therefore selected.

The condition on the observed latitude ensures rejecting measurements within ice covered region.

The condition on the land flag (oswLandFlag variable in the L2 product) ensures rejecting measurements acquired over land, not relevant for wave observation. As it is not specifically measured, it is only evaluated. The much larger number of wave mode acquisitions over land for S1A is explained by the acquisitions conducted for calibration purpose (over Africa, North and South America and Australia), not performed on S1B.

II.2.3.5 Retro-propagation

This is the first step of storm source identification. The required inputs are the peak period and direction of propagation given by the SAR L2 products and SWIM L2P products. From the observation time, the observation positions in the past are calculated following geodesic at group velocity in the direction opposite to the direction of propagation until land or ice is reached and for a maximum time of 14 days. Group velocity is estimated from the peak wavelength using linear wave dispersion relation, valid for swell with small steepness.
Along this past trajectory, the fact that the generation region is detected depends on two conditions. First, a minimum number of swell observations needs to converge on this region. This condition is referred to as *refocusing condition*. Second, the local sea surface wind speed needs to reach values compatible with the wind generation of the considered swell wavelength.
Indeed, a wave of peak period T_p requires a threshold wind speed $\widetilde{U}_{min}$ to be generated, such as,

$$\widetilde{U}_{min} = 0.12\ g\ T_p$$

where g is the Earth gravity. This ancillary wind information is given by the maximum 10-m wind speed indicated by ECMWF 0.125° wind model within 500 km of the swell propagated observation (+ a margin of 6% of the propagated distance since observation). In practice, because of the inability of input wind data to describe the highest wind speed range, **we consider a limit wind speed U_{min} equal to two-thirds of $\widetilde{U}_{min}$**. Thanks to this wind speed threshold, cross sea regions occurring under calm wind conditions are rejected from the following storm detection.

II.2.3.6 Refocusing and swell field definition

The space/time local maxima identified in the density maps defines the storm source. All the propagated swell observations in the identified regions are associated and define the new swell field. The generation time and region are the same for all observations.

From there, a refined analysis enables the detection of variations in the refocusing time and region with respect to the observation peak wavelength.

II.2.3.6.1 Peak period and peak direction estimation

Once the storm center is found, the swell propagated location, peak period and direction are estimated away from this origin until reaching the coast and following geodesics assuming linear wave propagation theory in the absence of currents, thus propagating at group velocity with fixed period. If no land is reached, the observations are propagated away from the storm for up to 10 days. The resulting characteristics are also known as pseudo-observations.

These pseudo-observations are given in the NetCDF file as a function of time (except for the peak period, which is constant). Before the generation by the storm and after land is reached, the peak direction, the longitude and latitude variables are replaced with their fill values (-999).

II.2.3.6.2 Significant wave height estimation

Considering a dissipation-free propagation, and in the absence of currents, continents and islands, the only phenomena modifying the swell energy or its significant wave height (H_s^{free}) are angular spreading, acting in transverse direction, and frequency dispersion, acting in radial direction. *Collard et al.* [2009] have demonstrated that far enough from the storm source and given the significant swell height at an angular great-circle distance α_0 from the storm source, its value could be estimated at any other position, α, as follows:

$$H_s^{free}(\alpha) = H_s(\alpha_0)\sqrt{\frac{\alpha_0 . \sin(\alpha_0)}{\alpha . \sin(\alpha)}}$$

Equation 1: Dissipation-free swell energy decay

This behaviour corresponds to an asymptotic trend, and it may not be well reproduced whenever too close to the storm source. In practice, for a 1,000 km-wide storm region, the threshold distance from the source for which this energy decay model is judged valid is 4,000 km. However, this limit is expected to vary depending on the storm region extension.

Experiments and measurements indicate that swell energy can dissipate during propagation [e.g. Snodgrass et al., 1966]. We follow the approach proposed by Collard et al. [2009] in which the swell energy decay is written using a linear decay parameter μ(m^{-1}) as follows:

$$\begin{aligned} H_s(\alpha) &= H_s^{\text{free}}(\alpha).\exp\left(-\mu\frac{\alpha-\alpha_0}{2}\right) \\ &= H_s(\alpha_0)\sqrt{\frac{\alpha_0.\sin(\alpha_0)}{\alpha.\sin(\alpha)}}\exp\left(-\mu\frac{\alpha-\alpha_0}{2}\right) \end{aligned}$$

Equation 2: Swell energy decay with a linear decay parameterization

For the decay parameter, we use a parameterization based on work conducted by Ardhuin et al. [2009], illustrated in Figure 4, and shown below:

$$\mu = a * \frac{H_{ss}}{W_p} + b$$

Equation 3: Parameterization of the linear decay parameter

where a = 4.e-5 and b = 2.e-7, H_{ss} is the significant wave height measured at 4,000 kmfrom the source, W_p is the peak wavelength.

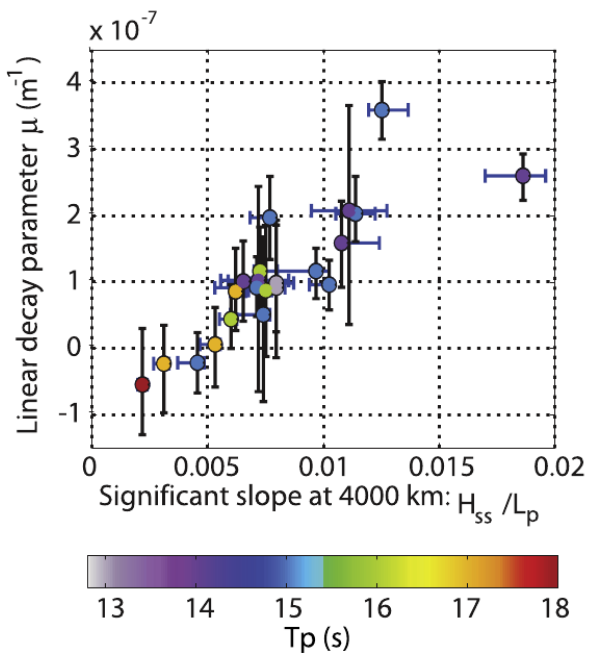

Figure 4 : Estimated linear dissipation rate (based on 22 swell events), as a function of the initial significant slope, ratio of the significant swell height and peak wavelength, H_s/L_p, taken 4,000 km from the storm center, for a variety of peak swell periods (colours). Figure taken from Ardhuin et al. [2009] to illustrate the decay variation

For each observation, the decay parameter is estimated by minimizing Equation 2 and Equation 3.

The evolution of the significant wave height with the swell propagation is given in the NetCDF file as a function of time (Table 8). Before the generation by the storm, after land is reached and closer to the source than 3,000 km, the significant wave values are replaced with their fill_values (-999). In the future, this 3,000 km limit could vary depending on the storm source size.

Table 8: Pseudo-observation variables and dimensions included in each L3 NetCDF file.

Variable name	Standard name	Long name	NetCDF Type	Units	Scale factor
time	time	Date of propagation in CNES convention	double	(days since 1950-01-01T00:00:00Z)	-
latitude	latitude	latitude of propagated observation	float	Degrees_east	-
longitude	longitude	longitude of propagated observation	float	Degrees_north	-
VAVH	sea_surface_wave_significant_height	Partition significant wave height along propagation	int	Meters	0.001
VPED	sea_surface_wave_from_direction_at _variance_spectral_density_maximum	Partition peak direction along propagation (met. convention)	int	Degrees	0.1
VTPK	sea_surface_wave_period_at_ variance_spectral_density_maximum	Partition peak period	int	Seconds	0.01

II.2.3.7 Quality flag estimation for the Level-3

Once the swell field is defined, the consistency between the swell propagated observations, or pseudo-observations, is used to reject observations which are afterwards defined as outliers. The procedure is described below:

- An *a priori* weighting, corresponding to a confidence index is estimated for each pseudo-observation, and for each different time step, from the generation time until 10 days after with a 3-hour time step. This weighting depends on:
    - The time between the propagated time and the observations: the further away from the observations time, the less confidence in the propagated information
    - The observation ambiguity factor: Extracted from the L2 input data, this is a confidence indicator in the 180° ambiguity on the propagation direction. The higher the ambiguity, the more confidence in the propagation direction
    - The azimuthal peak wavenumber with respect to the azimuthal cutoff: With larger azimuthal peak wavenumber with respect to the azimuthal cutoff, the swell partition is less affected by cut-off and associated energy attenuation.
    - The swell partition contrast: The partition contrast is defined as the ratio between the partition peak energy and the highest energy at the partition boundary. Swell partition with higher contrast is less sensitive to partitioning variations due to noise or multi-modal swell spectra.
- For every time step, the swell field spatial distribution of the three integral parameters is estimated together with the associated variance. This is performed using a surface fitting algorithm over a (r,θ) grid centered on the storm source.
- An iterative process is performed where pseudo-observations showing significant differences are rejected:

- For the significant wave height: pseudo-observations with differences larger than twice the relative variance are excluded.
- For the peak wavelength and peak direction: pseudo-observations with differences larger than a certain threshold are excluded. This threshold is estimated based on the storm source characteristics. The maximum size of the storm determines the maximum difference in wavelength and direction that can be observed at a given location from the storm source.

The swell observations identified as outliers by this iterative process have a lower quality flag attributed to either a lower quality of the input L2 observations or the hypothesis used for estimating the L3 product.

Variables related to the swell field observations are stored in a specific group called "obs_params" containing the variables described in Table 9 (below).

Table 9: Observation variables and dimensions included in each L3 NetCDF file for the obs_params group (continues on next page).

Variable name	Standard name	Long name	NetCDF Type	Units	Scale factor
obs_params/time	time	Date of propagation in CNES convention	double	(days since 1950-01-01T00:00:00Z)	-
obs_params/latitude	latitude	latitude of propagated observation	float	degrees_east	-
obs_params/longitude	longitude	longitude of propagated observation	float	degrees_north	-
obs_params/VAVH	sea_surface_wave_significant_height	Partition Significant Wave Height at observation time	int	Meters	0.001
obs_params/VPED	sea_surface_wave_from_direction_at_variance_spectral_density_maximum	Partition Peak Direction at observation time (met. convention)	int	Degrees	0.1
obs_params/VTPK	sea_surface_wave_period_at_variance_spectral_density_maximum_flag	Partition Peak Period	int	Seconds	0.01
obs_params/VAVH_flag	sea_surface_wave_significant_height_flag	Global Hs quality flag for each L2 swell partition (partition Hs coherency) - Values from 0 to 1	int	-	-
obs_params/VPED_flag	sea_surface_wave_from_direction_at	Global quality flag for each L2 swell partition (partition peak direction	int	-	-

Variable name	Standard name	Long name	NetCDF Type	Units	Scale factor
	_variance_spectral_density_maximum_flag	coherency) - Values from 0 to 5			
obs_params/VTPK_flag	sea_surface_wave_period_at_variance_spectral_density_maximum	Global quality flag for each L2 swell partition (partition peak period coherency) - Values from 0 to 5	int	-	-
obs_params/wave_spec	wave_spectrum	Estimated wave spectrum on log-polar grid	float	m^4	
obs_params/wavenumber_spec	wavenumber_spec	Array of the wavenumbers for polar spectra	float	rad/m	-
obs_params/direction_spec	direction_spec	Array of angular values for polar spectra	float	degrees from North, clockwise.	-
obs_params/partition_domain_spec	wave_partition	Estimated wave partition spectral domain on log-polar grid	int	-	-
obs_params/resolution_spec	resolution_spec	Azimuth cut-off wavelength as function of wave direction, for the input L2 spectrum	float	m	-
obs_params/L2_partition_quality_flag	L2_partition_quality_flag	Quality flag per partition for the input L2 spectrum	int	-	-

III VALIDATION FRAMEWORK

III.1 Overview

The validation aims to check the quality of the produced significant wave height and the performances of the key processing steps. Different points are assessed by the validation task:

- The data availability and spatial/temporal coverage,
- The data quality flag monitoring,
- The performance monitoring using cross-overs,

It should be noted, that while the two first points are shortly presented in this document, their results are only made available on CLS monitoring website [CLSMonitor]. Table 10 lists the different metrics, following the nomenclature applied by the Copernicus Marine Service, that are used by the Production Unit to validate the products and shortly described in the following subsections. This document focuses on the quality of the main integral parameters used to describe the swell fields: significant wave height (Hs), peak wavelength (Wl), and peak direction (Dp). These parameters are generically referred to as <SPC_params> in Table X.

The validation of these parameters consists of comparing the integral wave parameters obtained from one satellite configuration with:

- another satellite or satellite configuration in the SPC-FWK L3 product (internal validation)
- In-situ data from National Data Buoy Center (NDBC)

In addition to the statistical comparison, some additional validations were previously carried out as case studies. Interested readers are invited to refer themselves to Wang et al. (2018) for further discussions about investigation of uncertainties impact at different steps of the processing.

N.B.: Link between Peak period (Tp) and Peak Wavelength (Wl)

The relation between period and wavelength is given by the dispersion relation. In linear wave theory, when considering waves in the deep water (i.e. with no bathymetry effect) this relation writes,

$$W_l = \frac{g}{2\,\pi}\,T_p$$

Where g is the acceleration by gravity. Therefore, both parameters are usually used interchangeably depending on the user. In the following, the results are given in terms of Wl.

Table 10: List of metrics for WAVE-TAC SPC-derived measurements.

Name	Description	Ocean parameter	Supporting reference dataset	On Monitoring website	Results on this document
SPC_L2-NC-AVAIL-<period> SPC_L2-NC-USED-<period>	Number of input SPC measurements (swell partitions) available/used	Swell partitions	None	Yes	No
SPC-L2-NC-SAT-AVAIL-GLB-M	Maps of available input swell partitions over a month	Swell partitions	None	Yes	No
SPC_L3-NC-VALID-<period> SPC_L3-NC-REJ-<period>	Number of SPC swell partitions valid/rejected by the SPC-FWK	Swell partitions	None	Yes	No
SPC-L3-NC-SAT-VALID-GLB-M SPC-L3-NC-SAT-REJ-GLB-M	Maps of swell partitions valid / rejected by the SPC-FWK over a month	Swell partitions	None	Yes	No
<SPC_params>_L3-NC-SAT-BIAS_T-XOVER <SPC_params>_L3-NC-SAT-RMSD_T-XOVER <SPC_params>_L3-NC-SAT-SI_T-XOVER	Bias / Root-Mean Square Deviation (RMSD) / Scatter Index (SI) between satellite observations at crossover positions, for various mono- and multi-missions configurations.	Significant wave height (Hs), peak wavelength (Wl) and peak direction (Dp)	None	Yes	Yes
<SPC_params>_L3-CLASS2-MOOR-BIAS <SPC_params>_L3-CLASS2-MOOR-RMSD <SPC_params>_L3-CLASS2-MOOR-SI	Integral parameters differences between satellites and In Situ buoys	Significant wave height (Hs), peak wavelength (Wl) and peak direction (Dp)	NDBC (National Data Buoy Center) spectral wave buoys	Yes	Yes

III.2 Metrics description

This subsection describes the metrics listed in Table X in more details.

III.2.1 Input data availability and spatio-temporal coverage

The metrics SPC_L2-NC-AVAIL-<period> and SPC_L2-NC-USED-<period> are used to quantify the number of input L2 products, giving the number of waves partitions in total and the number of wave partitions used for the L3 production.

The associated SPC-L2-NC-SAT-AVAIL-GLB-M and SPC-L2-NC-SAT-USED-GLB-M represent monthly global maps of SPC_L2-NC-AVAIL-<period> and SPC_L2-NC-USED-<period> over a 2° x 2° grid.

The spatial coverage of the measurements is monitored on a daily/weekly/monthly basis for S1A, S1B and SWIM and is made available on the CLS monitoring website [CLSMonitor].

III.2.2 Data quality flag monitoring

The metrics SPC_L3-NC-VALID-<period> and SPC_L3-NC-REJ-<period> are used to quantify the swell partitions flagged as good quality or rejected over the SPC-FWK process.

The associated SPC-L3-NC-SAT-VALID-GLB-M and SPC-L3-NC-SAT-REJ-GLB-M represent monthly global maps of SPC_L3-NC-VALID-<period> and SPC_L3-NC-REJ-<period> over a 2° x 2° grid.

The temporal evolution of these metrics is given monitored on a daily/weekly/monthly basis for S1A, S1B and SWIM and is made available on the CLS monitoring website [CLSMonitor].

III.2.3 Performance monitoring using cross-overs

The performance monitoring of the L3 products is performed both:

- for all swell measurements: swell observations and pseudo-observations, and
- for good quality swell measurements only, i.e. partitions that are 'refocusing' in the SPC-FWK L3 process only.

It is performed for the three main swell parameters: Hs, Wl and Dp.

III.2.3.1 <u>Between satellites</u>

The metrics <SPC_params>_L3-NC-SAT-BIAS_T-XOVER, <SPC_params>_L3-NC-SAT-RMSD_T-XOVER and <SPC_params>_L3-NC-SAT-SI_T-XOVER quantify the bias, Root-Mean Square Deviation (RMSD) and Scatter Index (SI) between two Hs (or two Wl or two Dp) at crossover positions.

Various configurations of crossovers are considered:

- WV1 vs WV2: 2 configurations for each S1 satellite, ascending or descending,
- Pairwise comparison between S1 satellites: 4 configurations for WV1 and WV2, both ascending or both descending,
- Swell observed in SPC range vs. azimuth directions:2 configurations per S1 satellite (WV1 and WV2) and 1 configuration for SWIM.
- S1 satellites vs SWIM, both ascending or both descending
- SWIM left / SWIM right looking: 1 configuration

These comparisons along with the geolocation maps of the identified cross-over (2D density maps) are made available on the CLS Monitoring Website [CLSMonitor] and updated several times a month.

III.2.3.2 Against in-situ buoys

The metrics <SPC_params>_L3-CLASS2-MOOR-BIAS, <SPC_params>_L3-CLASS2-MOOR-RMSD and <SPC_params>_L3-CLASS2-MOOR-SI quantify the bias, Root-Mean Square Deviation (RMSD) and Scatter Index (SI) between Hs (or Wl or Dp) obtained from SPC-FWK L3 and Hs (or Wl, or Dp) obtained from buoys.

To do these comparisons the buoys' raw data obtained on NDBC website needs to go through the following steps:

1. Buoy directional wave spectra reconstruction using the Maximum Entropy Method (MEM, Donelan et al. 2015) ,
2. Buoy directional wave spectra partitioning (using a watershed algorithm)
3. Cross-assignment with SPC-FWK L3 products closest wave partitions.

In the future, this processing is expected to be improved using an enhanced filtering of the buoy swell partitions using time continuity criteria for swell partition belonging to the same swell field as done for the CFOSAT partitions.

These comparisons are made available on the CLS Monitoring Website [CLSMonitor] and updated several times a year.

III.2.3.3 Crossover position

Different types of crossover are performed depending on whether we account :

- for all input (or used) partitions, known as L2 crossovers
- only for 'refocusing' partitions, known as L3 crossovers

L2 crossovers correspond to all used points that are located less than 200 km from one another when one of the two observations is propagated for a maximum threshold time:

- Values for the Hs corresponds to the Hs at observation point and the threshold time equals 18h (the Hs value cannot be propagated from the L2 without knowing the storm source).
- Values for the peak wavelength and direction corresponds to the propagated values, for a maximum propagation time of 48h.

L3 crossovers correspond to all 'refocusing' points that are located less than 200 km from one another when one of the two observations is propagated for a maximum 48h. This is the same for all parameters.

IV VALIDATION RESULTS

This section presents an overview of the performance of the SPC-FWK L3 product in terms of swell integral parameters. It consists in two subsections, the first one about the internal validation of the product with comparisons between satellites and the second one about the validation against moored in-situ buoys.

IV.1 Comparisons between satellites

Amongst the satellite comparison configurations listed in Section III.2.3.1, this document only presents two of them as an example (the rest can be found in the CLS Monitoring Website [CLSMonitor]) in the following subsections:

- Sentinel-1A WaVemode 1 (S1A WV1) vs Sentinel-1A WaVemode 2 (S1A WV2)
- Sentinel-1A vs SWIM

IV.1.1 S1A WV1 vs S1A WV2

Figure 5 shows the resulting scatter plots of the comparison between S1A WV1 and S1A WV2, while Figure 7 shows the maps of the crossovers' associated locations.

The comparison is done both for L2 crossovers where all input partitions are considered (see Figure 5.a, Figure 5.c, and Figure 5.e) and for L3 crossovers where only the partitions 'refocusing' by the SPC-FWK L3 algorithm are considered (see Figure 5.b, Figure 5.d and Figure 5.f). The associated locations for L2 crossovers are shown in Figure 6.a for Hs and Figure 6.c for Wl and Dp; while the associated location for L3 crossovers are shown in Figure 6.b for all parameters.

The results are given for the three parameters : Hs (Figure 5.a and Figure 5.b), Wl (Figure 5.c and Figure 5.d) and Dp (Figure 5.e and Figure 5.f).

Two stricking observations can be made:

- There are much less points for the L3 comparisons than for the L2, which was to be expected since not all inputs points 'refocused'. Here we see around 14% of Wl and Dp points after 'refocusing' selection (if we consider the Hs points this number is higher as not all inputs Wl and Dp points have valid Hs).
- The 'refocusing' selection dramatically enhances the performance between S1A configurations, with a RMSD decrease from 98m to 26m for Wl and from 83° to 8° for Dp. This improvement of the performance also exists for Hs to a smaller extent, with a RMSD decrease from 1.15m to 0.64m. This is to be expected as the Hs parameter is the more complicated to obtain and ascertain for swell partitions.

To go further into details it can be seen that the biases remain approximately the same for the peak direction and significant wave height when comparing the L2 and the refocused propagated observations. The peak wavelength however shows a bias improvement from -14m to -1.3m with the refocused data.

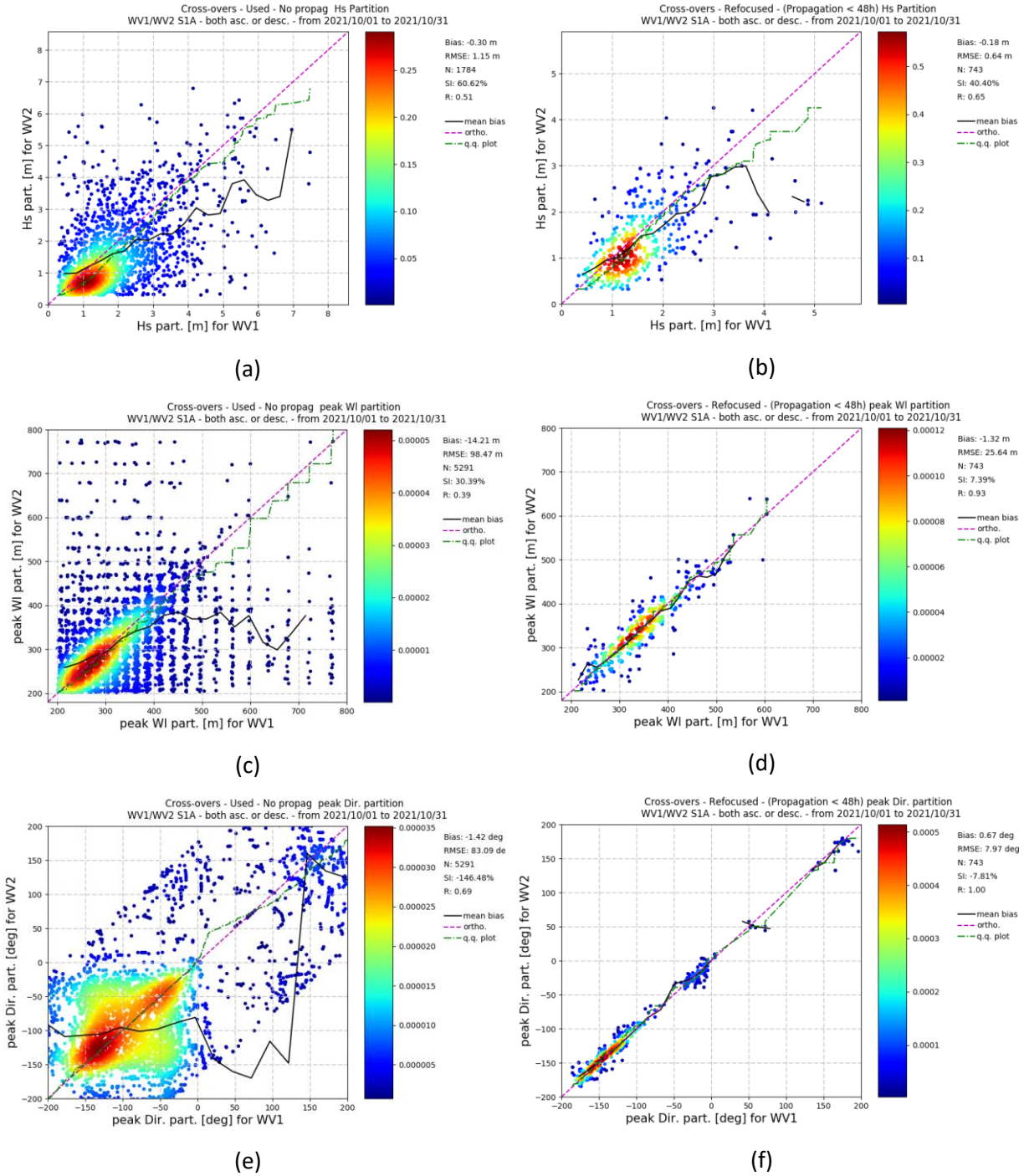

Figure 5: Comparisons between acquisitions in WV2 vs. WV1 of Sentinel-1A at (left – a,c,e) L2 cross-over points and at (right – b,d,f) L3 crossover points. Scatter plots show the comparisons for Hs (top – a, b), peak wavelength (middle – c, d) and peak direction (bottom – e, f) over the month of October 2021. See Figure 7 for the location of the points.

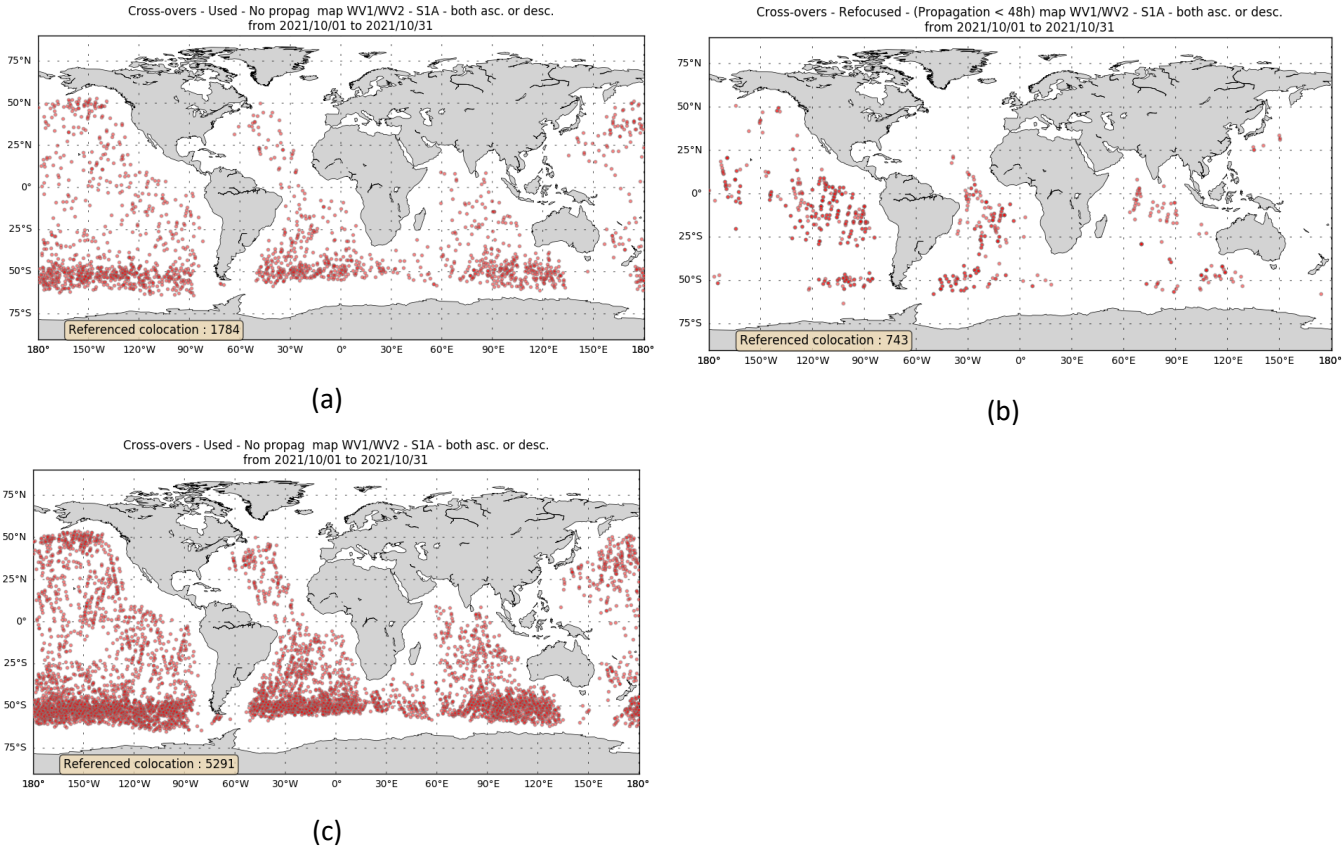

Figure 6: Location of the (left - a,c) L2 cross-over points and (right – b) L3 crossover points for acquisitions in WV2 vs. WV1 of Sentinel-1A used in scatterplots in Figure 5. While the L3 crossovers are the same for all three parameters (Hs, Wl and Dp), the L2 crossovers for Hs (a) are computed with a maximum time of 18h while for both Wl and Dp (c) the maximum propagation time is 48h.

IV.1.2 S1A vs CFOSAT/SWIM

Figure 7 shows the scatter plots of the comparison between S1A and CFOSAT/SWIM for L3 crossovers, along with the associated locations of these crossovers (Figure 7.d).

The results are given for the three parameters : Hs (Figure 7.a), Wl (Figure 7.b) and Dp (Figure 7.c).

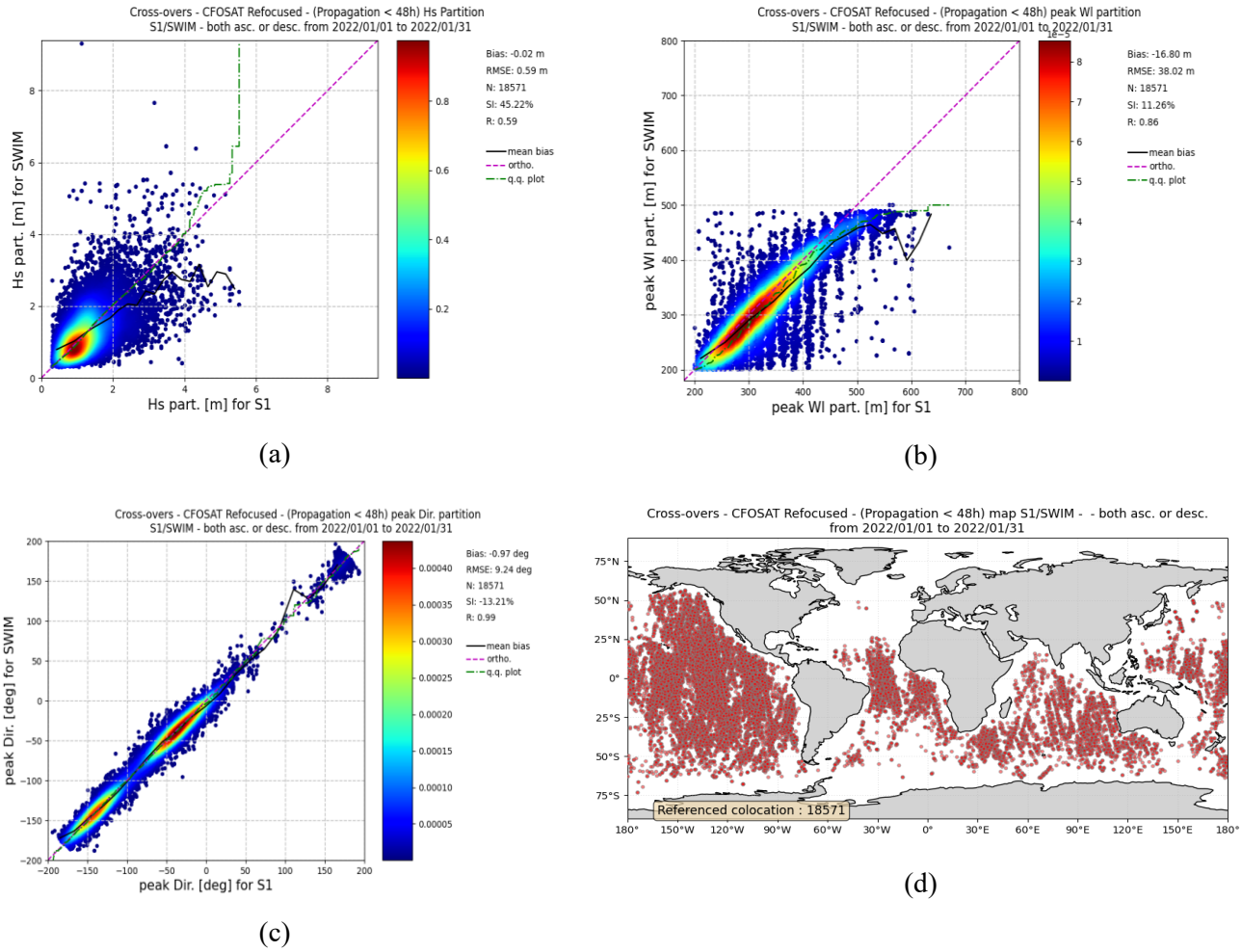

Figure 7: Comparisons between L3 products for acquisitions from SWIM vs. S1 at cross-over points for Hs (top left - a), peak wavelength (top right - b) and peak direction (bottom left - c) for the month of January 2021. Bottom right (d) shows the location of the cross-over points used in these scatterplots for all L3 products (maximum propagation time of 48h).

The agreement between S1A and CFOSAT/SWIM is of the same order of magnitude as the agreement within S1A configurations (see previous subsection) with RMSD values of:

- 0.59m for Hs
- 18 m for Wl, and
- 9° for Dp

The major differences can be found in the peak wavelength estimation, where the CFOSAT/SWIM cutoff at 500m wavelenght leads to underestimation for the high wavelenght compare to S1A. It should also be noted that while the bias between CFOSAT/SWIM and S1A has been reduced with the new calculation

of the peak wavelength for CFOSAT/SWIM (the integral parameters are now computed over the heigth spectrum, as for S1A), the remaining bias of 16m is still consequent and may be linked to the type of acquisition.

IV.2 Against in-situ buoys

In this section, the results of the validation against in-situ buoys are shown for:

- S1A WV1 in Figure 8.a and Figure 8.b,
- S1A WV2 in Figure 8.b and Figure 8.d, and
- CFOSAT/SWIM in Figure 8.e and Figure 8.f

For each satellite configuration, the comparisons are shown both for Wl (Figure 8.a, Figure 8.c and Figure 8.e) and for Hs (Figure 8.b, Figure 8.d and Figure 8.f). Looking at the spaceborne vs in situ comparisons, it can be seen that:

<u>For Hs:</u>

- for S1A: WV1 and WV2 (top and middle panels, right – b, d): mean bias for partitions Hs of 8 cm and –3 cm, respectively. The scatter index is between 50 and 60%
- for SWIM (bottom panels, right - f): Mean bias of -13 cm and the scatter index is of 60%. It should be noted that there, the number of colocations with buoys is ca. 7 times greater than for S1A-WV1 and S1A-WV2.

All three configurations show an overestimation of Hs compared to the buoy for buoy Hs ranging from [0.5 m, 2 m].

<u>For Wl:</u>

- for S1A: WV1 and WV2 (top and middle panels, left – a, c): mean bias for partitions Wl of 22 m and 15 m, respectively. The scatter index is between 15 and 25%. Both configurations show a slight overestimation of Wl compared to the buoy Wl between 250m and 350m of buoy Wl.
- for SWIM (bottom panels, left - e): Mean bias of –2m and larger for Wl > 500 m due to SWIM max. Wl. Limitation. The scatter index of 20% is consistent with the one for the S1A configurations.

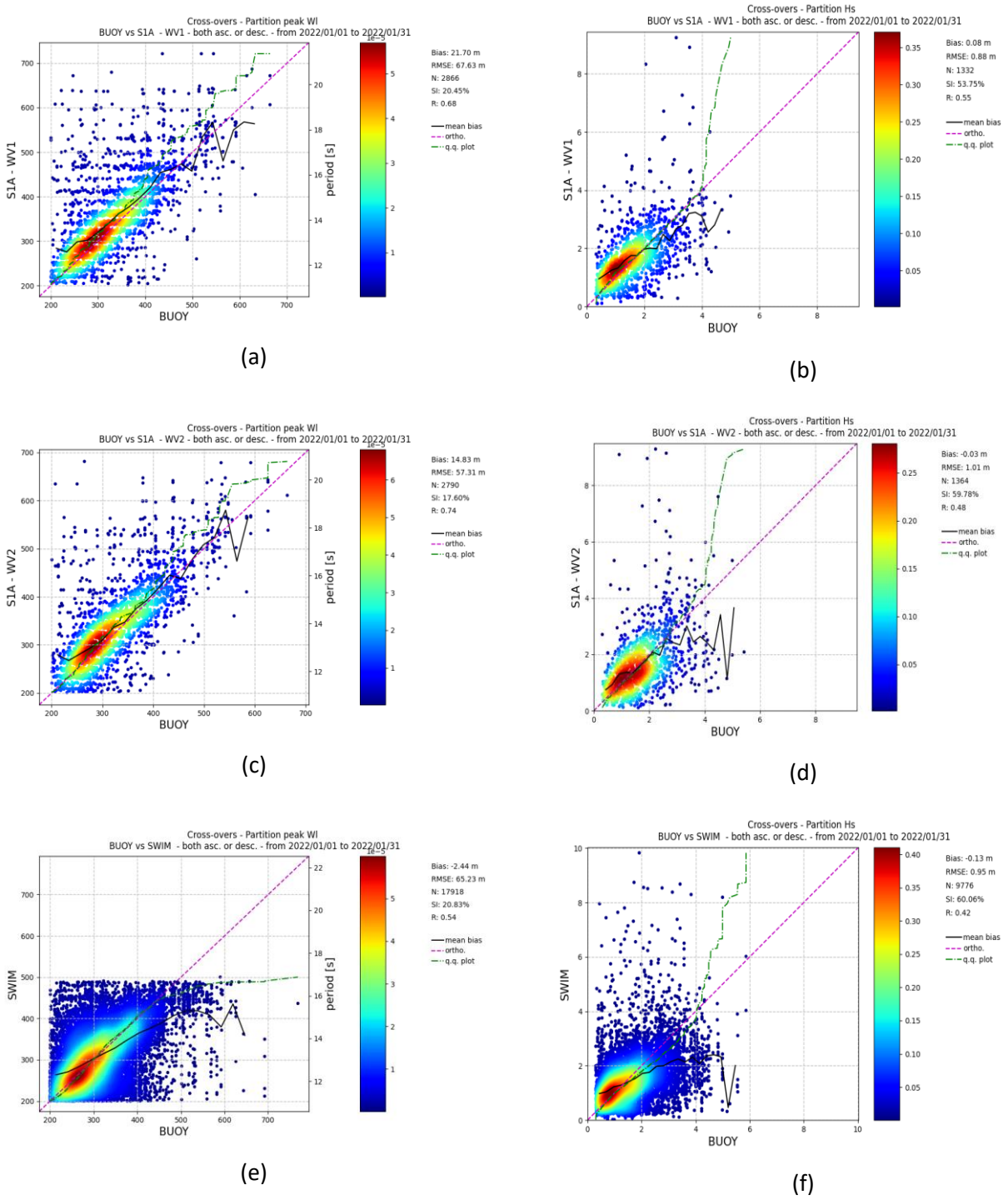

Figure 8: Comparisons between L3 products peak wavelength (left – a, c, e) and Hs (right – b, d, f) for acquisitions from S1A-WV1 (top – a, b), S1A-WV2 (middle – c, d) and SWIM (bottom – e, f) vs. NDBC buoys using cross-over over the month of January 2021.

IV.3 Validation Conclusion

Impact on selecting refocused data (comparisons between Sentinel-1 configurations):

With respect to the overall set of Level-2 (L2) swell observations used in the L3 analysis (i.e. which have passed the selection criteria detailed in Table 7), 14% of the L2 input swell observations are kept in the L3 CMEMS spectral products for internal comparison.

The biases remain approximately the same for the peak direction and significant wave height when comparing the inputs and the refocused propagated observations. The peak wavelength, however, shows a bias improvement from -14m to -1.3m with the refocused data.

The RMSD (Root Mean Square Difference) is greatly improved for both:

- From 98 m to 26 m for the peak wavelength,
- From 83° to 8° for the peak direction,

Comparison S1A vs CFOSAT/SWIM

The major differences can be found in the peak wavelength estimation, where the CFOSAT/SWIM cutoff at 500m wavelenght leads to underestimation for the high wavelenght compare to S1A.

Overall, thanks to the many co-locations provided by cross-overs for different comparisons (SWIM vs. S1, S1 and SWIM vs. Buoy) we can conclude that the strong under-estimation of SWIM peak wavelength estimated from slope spectrum has been reduced by considering height spectrum from –19 m to –2 m for SWIM/buoy comparisons and from –30 m to –16 m for S1A/SWIM comparisons.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to track and describe changes that may occur in the current operational system: due to outages, version (see Table 11) or upstream data changes (e.g. addition or loss of a satellite).

V.1 System version changes

Table 11: L3 spectral wave chain version changes.

System version	QUID version	Date of Entry in Service of the change	Description of the change	Impact on product quality?
1.0	1.0	03.07.2018	Entry into Service of the L3 spectral wave chain with Sentinel-1A and Sentinel-1B missions	Yes
1.5.16	1.3	14.12.2021	Addition of CFOSAT SWIM off-nadir measurements	Yes
1.5.24	1.3	04.04.2022	Bug correction on scaling factor application (it was applied twice before)	Yes
2.0.23	1.3	30.11.2023	Code update to update from Python 2 to Python 3 Improvement: Add swim statistics on the monitoring website: satwave-report.cls.fr/ Improvement: Activate CFOSAT/SWIM Data in L4 NRT Fix: Fill values might not be correctly written in some L3 for VPED, VAVH and VTPK variables.	Yes
2.1.4	1.3	11.07.2024	EIS June 2024: Modification of dissemination process (change in MDS address)	No
2.1.5	1.4	13.11.2024	Change filenames for L3 obs	No
2.1.6	1.4	12.12.2024	Bug fix: related to not applicable peak-to-boundary ratio	Yes
2.1.9	1.4	27.02.2025	Add delivery detection time (for KPI)	No
2.1.10	1.4	27.02.2025	Change to deal with potential corrupted input S1A files	No
2.1.11	1.4	27.02.2025	Change to read new input L2P CFOSAT (size modification only)	No
3.0.0	1.5	04.11.2025	Refactoring of the code to allow for other input types. Enhancement of input filtering for CFOSAT to use more data.	Yes

V.2 Main constellation events impacting data availability and quality

Different events can lead to a change in data availability and quality. Such events are usually:

- A change in the SAR constellation: the loss or introduction of a SAR in the constellation directly impacts the number of SAR measurements available.
- For a specific platform, a reduction of the number of SAR measurements available in input of the L3 spectral wave system processing. This can be linked with an anomaly on board the platform or on the ground segment, preventing the data reception and impacting the L0 to L2 processing. It can also be induced by an abnormal acquisition by the L3 spectral wave system.
- A change in the L0 - L2 processing can lead to changes in the quality of the measurements in input of the L3 spectral wave system processing. For example, new versions of L2 upstream products are regularly released, to account for corrections and developments of this upstream processing.

Currently, 3 SAR missions constitute the SAR constellation available in NRT.

- Sentinel-1A (S1A) is the oldest in the constellation.
- Sentinel-1B (S1B), but end of mission occurred on 23.12.2021
- CFOSAT (CFO).

Table12 summarizes the main events affecting the data availability in NRT conditions.

Table12: Main events affecting the data availability in NRT conditions.

Date	Platform	Event
22.04.2018	S1A, S1B	Introduction of Sentinel-1A and Sentinel-1B missions in the L3 wave spectral chain
14.12.2021	*CFO*	Introduction of CFOSAT mission in the L3 wave spectral chain
23.12.2021	S1B	End-of Mission, official since 03.08.2022[7]

V.3 Historical changes in the calibration and potential impact for users

No change to report since the start of the service.

[7] https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-1/Mission_ends_for_Copernicus_Sentinel-1B_satellite

VI QUALITY CHANGES SINCE PREVIOUS VERSION

Since EIS Nov 2025, the filtering of the input CFOSAT data has changed, allowing the consideration of more partitions for the swell analysis (within 0.3 and 1 m of HsTable 6) as shown in Table 6.

In Table 13 we compare the Estimated Accuracy Numbers (EANs) for the previous version and for the new version. The data format has not been modified.

Table 13: Estimated Accuracy Numbers (EANs) for Significant wave height, peak wavelength and peak direction, computed for both previous and new version from 01.01.2022 to 31.01.2022.

			Previous version			New version		
Significant Wave Height	**Comparison between**		**Bias**	**RMSD**	**SI**	**Bias**	**RMSD**	**SI**
	x-axis	**y-axis**	**(m)**	**(m)**	**(%)**	**(m)**	**(m)**	**(%)**
	L3 from S1	L3 from CFOSAT	0.07	0.69	45.04	-0.02	0.59	45.22
	NDBC buoys	L3 from CFOSAT	0.11	0.95	54.06	-0.13	0.95	60.06
Peak wavelength	**Comparison between**		**Bias**	**RMSD**	**SI**	**Bias**	**RMSD**	**SI**
	x-axis	**y-axis**	**(m)**	**(m)**	**(%)**	**(m)**	**(m)**	**(%)**
	L3 from S1A - WV2	L3 from CFOSAT	-24.61	36.46	11.43	-16.80	38.02	11.26
	NDBC buoys	L3 from CFOSAT	-19.99	51.34	16.45	-2.44	65.23	20.83
Peak direction	**Comparison between**		**Bias**	**RMSD**	**SI**	**Bias**	**RMSD**	**SI**
	x-axis	**y-axis**	**(°)**	**(°)**	**(%)**	**(m)**	**(m)**	**(%)**
	L3 from S1A - WV2	L3 from CFOSAT	-0.32	10.07	-12.66	-0.97	9.24	-13.21
	NDBC buoys	L3 from CFOSAT	0.79	35.58	13.21	-1.20	61.53	22.97

VII REFERENCES

Ardhuin, F., Chapron, B., and Collard, F. “Observation of Swell Dissipation across Oceans.” Geophys. Res. Lett 36 (2009).

Collard, F., Ardhuin, F., and Chapron, B. “Monitoring and Analysis of Ocean Swell Fields Using a Spaceborne SAR: A New Method for Routine Observations.” J. Geophys. Res. 114 (2009): C07023.

Donelan, M., Babanin, A., Sanina, E., & Chalikov, D. (2015). A comparison of methods for estimating directional spectra of surface waves. Journal of Geophysical Research: Oceans, 120(7), 5040–5053. http://onlinelibrary.wiley.com/doi/10.1002/2015JC010808/full

Heimbach, P., and Hasselmann, K. “Development and Application of Satellite Retrievals of Ocean Wave Spectra.” Elsevier Oceanography Series 63 (2000): 5–33.

Holt, B., Liu, A. K, Wang, D.W. Gnanadesikan, A., and Chen, H. S. “Tracking Storm-Generated Waves in the Northeast Pacific Ocean with ERS-1 Synthetic Aperture Radar Imagery and Buoys.” Journal of Geophysical Research 103, no. C4 (1998).

Husson, R. “Development and Validation of a Global Observation-Based Swell Model Using Synthetic Aperture Radar Operating in Wave Mode.” IUEM - EDSM, 2012.

Munk, W. H.,. Miller, G.R., Snodgrass, F.E. and Barber, N. F. “Directional Recording of Swell from Distant Storms.” Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences 255, no. 1062 (1963): 505–584.

Snodgrass, F. E., Hasselmann, K. F., Miller, G. R., Munk, W. H., & Powers, W. H. (1966). Propagation of ocean swell across the Pacific. Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences, 259(1103), 431-497.

Wang, H., Mouche, A., Husson, R., & Chapron, B. (2018, July). Dynamic validation of ocean swell derived from Sentinel-1 wave mode against buoys. In IGARSS 2018-2018 IEEE International Geoscience and Remote Sensing Symposium (pp. 3223-3226). IEEE.

Wang, X., Husson, R., Jiang, H., Chen, G., and Gao, G. “Evaluation on the Capability of Revealing Ocean Swells from Sentinel-1A Wave Spectra Measurements.” Journal of Atmospheric and Oceanic Technology 37, no. 7 (2020): 1289–1304.

Wang H., Mouche A., Husson R., Grouazel A., Chapron B., Yang J.(2022). Assessment of Ocean Swell Height Observations from Sentinel-1A/B Wave Mode against Buoy In Situ and Modeling Hindcasts. Remote Sensing, 14(4), 862 (20p.). Publisher's official version: https://doi.org/10.3390/rs14040862, Open Access version: https://archimer.ifremer.fr/doc/00751/86278/

Wang, H., Mouche, A., Husson, R., Chapron, B., Yang, J., Liu, J., & Ren, L. (2022). Quantifying Uncertainties in the Partitioned Swell Heights Observed From CFOSAT SWIM and Sentinel-1 SAR via Triple Collocation. IEEE Transactions on Geoscience and Remote Sensing, 60, 1–16. https://doi.org/10.1109/TGRS.2022.3179511